

Encrypting a Hard Drive

Cybersecurity Portfolio Project

1. Project Overview

This project focused on protecting sensitive data using BitLocker Drive Encryption in Windows. The project demonstrates how full disk encryption helps secure files and prevent unauthorized access to data stored on a system.

2. Objective

The goal of this project was to explore and configure BitLocker Drive Encryption to improve endpoint security and strengthen data protection on a Windows system.

3. Tools Used

- Windows 10
- BitLocker Drive Encryption
- Control Panel
- VirtualBox

4. Skills Demonstrated

- Data Protection
- Disk Encryption
- Endpoint Security
- Windows Administration
- Security Hardening
- Access Control

5. Lab Environment

The lab environment was created using a Windows 10 virtual machine running inside VirtualBox. BitLocker Drive Encryption settings were accessed through the Windows Control Panel.

6. Steps Performed

1. Opened the Windows Control Panel.
2. Navigated to:
 - System and Security
 - BitLocker Drive Encryption
3. Reviewed the operating system drive encryption settings.
4. Selected the option to turn on BitLocker.
5. Examined available encryption options and drive protection settings.
6. Documented the encryption process and security features.

7. Screenshots

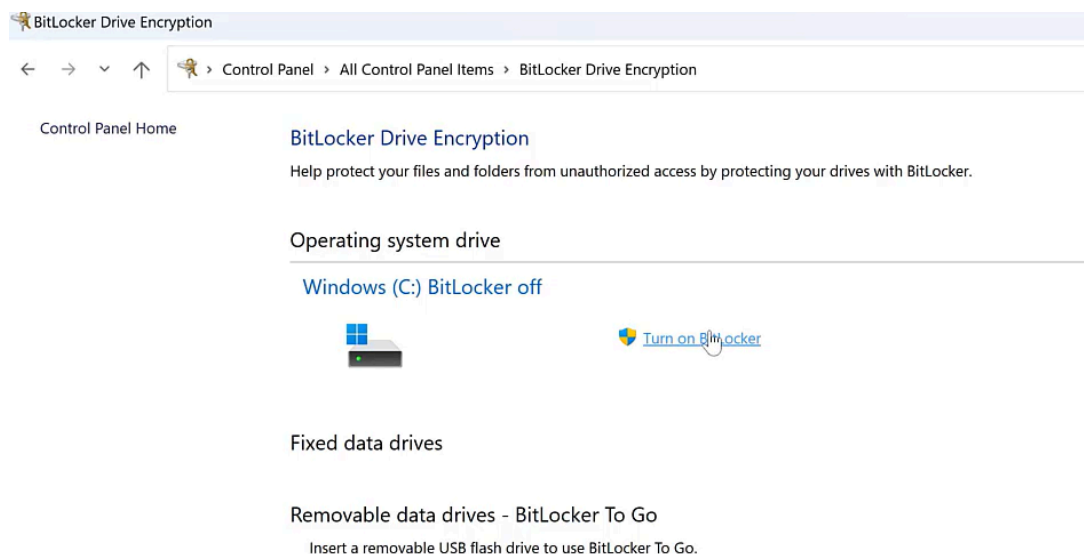
Figure 1 — Accessing BitLocker Drive Encryption

- Opened BitLocker settings from the Windows Control Panel.



Figure 2 — BitLocker Drive Encryption Settings

- Viewed operating system drive protection settings.
- Selected “Turn on BitLocker” for the Windows drive.



8. Security Benefits

Full disk encryption protects sensitive information by preventing unauthorized users from accessing files stored on a device.

BitLocker helps:

- Protect lost or stolen devices
- Secure sensitive data
- Improve compliance
- Reduce unauthorized access risks

9. What I Learned

I learned how Windows BitLocker provides full disk encryption and strengthens endpoint security through drive-level protection.

I also gained experience navigating encryption settings and understanding how encryption protects stored data.

10. Conclusion

This project demonstrated practical cybersecurity skills related to data protection, Windows encryption technologies, and endpoint security using BitLocker Drive Encryption.